

Firebolt 9 Crypto API Specification

Document status	DRAFT
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Note: this is a draft of the latest Firebolt spec

API version	Link	Summary
internal draft		

Interpretation

This page is an abstract, language-independent definition of the Firebolt Core API. The following terms are used:

	Term	Meaning
1	enum	an instance of a type that can have one of a pre-defined set of values. Typically represented by a string in JSON-RPC, but an integer in C++
2	list of	an ordered collection of values
3	list of... unordered	an unordered collection of values
4	one of	an instance of a primitive type, such as unsigned or string, that can have one of a pre-defined set of values. Not an enum
5	monotonic	a numeric value that is greater than the last instance of that attribute
6	generic error	an error that can be returned by any method call, documented in the Errors section
7	specific error	an error that can be returned by specific methods, documented along with the method

Key

Color	Meaning
Green	Approved - ready for development
Red	Not approved - not ready for development

Types

	Type	Definition
1	unsigned	32-bit unsigned integer
2	double	64-bit double-precision floating point number

Errors

	ID	Error	Notes
Specific	-xxxxa	Operation failed	Includes any resourcing errors, operation not allowed / supported, any internal or hardware error

-xxxxb	File to be loaded is not available or corrupted		
-xxxxc	File name does not represent valid key name		
-xxxxd	Invalid key type	xxx	
-xxxxe	Invalid key		
-xxxxf	Session to be reinitialized		The vault session is invalidated. It needs to be reinitialized before next usage.
-xxxxg	Invalid Parameters		Null or Invalid parameters
-xxxxh	Authentication Failed		Operation not permitted for specific app credentials - example, when an app tries to use another apps credentials
-xxxxi	Invalid Init Data		Length of IV or nonce supplied as parameter is not equal to the expected length
-xxxxj	Data size exceeded		
-xxxxk	Key Provision Error		
-xxxxl	Too small output buffer		
-xxxxm	Sealed Key Cannot Export Error		

Methods

	Module	Method	Parameters	Returns	Specific errors	Firebolt global	C++	JS	Description
1	Crypto	generateKey	- keyType - enum - AES128 - AES256 - HMAC128 - HMAC160 - HMAC256 - keyUsage - list of enums, unordered - optional - ENCRYPT - DECRYPT - SIGN - DERIVE - UNWRAP	keyId - unsigned	-xxxxa - Key Creation Error -xxxxe - Invalid File name -xxxxh - Authentication failure -xxxxd - Invalid key type	Yes	Yes	No	Generate symmetric algorithm key file, load it into key store and provide handle for further access. Generated key this way shall be considered local to the application generating it - shall not be shared. Not persisted during this call. To use exportKey in case persistence is required.
2	Crypto	generateKeyPair	- algorithm - enum - AUTH_DH Optional parameter for AUTH_DH algo: <pre>{ • primePDataOffset - unsigned • primePDataLength - unsigned • generatorGDataOffset - unsigned • generatorGDataLength - unsigned }</pre> Optional parameter for RSA algo (not supported, just for ex): <pre>{ • modulusBits - unsigned • publicExponent - unsigned }</pre>	- privateKeyId - unsigned - publicKeyId - unsigned	-xxxxh - Authentication failure -xxxxa - Operation failed -xxxxz / Invalid Parameters	Yes	Yes	No	Create authenticated DH session keys with algorithm "AUTH_DH" Authenticated Diffie-Hellman. Mapping from Netflix APIs: <pre>generatorG; // from base_buf primeP; // from modulus</pre>

3	Crypto	loadKey	- keyName - string, Locator that points to a key that is pre-provisioned - provisionType - enum - optional - pre-provisioned (global)	unsigned - keyId (Reference to the loaded key)	-xxxxb - Not available or corrupted -xxxxc - Does not represent valid key -xxxxh - Authentication failure	Yes	Yes	No	Load a pre provisioned key by locator
4	Crypto	unloadKey	keyId - unsigned	None	-xxxxh - Authentication failure -xxxxa - Operation failed	Yes	Yes	No	Delete key from key store (may mean unload key from TA if used).
5	Crypto	keyExists	keyName - string	- bool - unsigned - keyId	-xxxxh - Authentication failure	Yes	Yes	No	Returns true if a valid key exists (loaded to be used) with the key name. Returns false otherwise.
6	Crypto	getKeyInfo	keyId - unsigned	- keyType - enum - AES128 - AES256 - HMAC128 - HMAC160 - HMAC256 - keyUsage - enum - ENCRYPT - DECRYPT - SIGN - DERIVE - UNWRAP	-xxxxe - Invalid key -xxxxh - Authentication failure	Yes	Yes	No	Get key info / size from keystore

				- keyAlgo - enum - AES - CBC - AES - CTR - DIGEST - SHA256 - DIGEST - SHA384 - DIGEST - SHA512					
7	Crypto	flushKeys	None	None	-xxxxh - Authentication failure	Yes	Yes	No	Unloads all the keys loaded until now. They cannot be used again. Returns INVALID_KEY error on further key usage.
8	Crypto	importKey	- keyType - enum - AES128 - AES256 - HMAC128 - HMAC160 - HMAC256 - Import/Export Type - Protected - Clear - Key blob to be imported - device-bound, integrity-protected, confidentiality-protected blob	keyId - unsigned (Reference to the loaded key)	-xxxxh - Authentication failure -xxxxk - Key Provision Error			No	Clear import and export is not supported by Comcast / Sky implementation
9	Crypto	exportKey	- keyId - unsigned - Import/Export Type - enum - Protected - Clear	keyBlob - list of unsigned, ordered	-xxxxh - Authentication failure -xxxxm - Sealed Key Cannot Export Error -xxxxe - Invalid key	Yes	Yes	No	Returns device-bound, integrity-protected, confidentiality-protected blob May have the metadata export based on the implementation.

10	Crypto	sign	- algorithm - enum - HMAC - keyId - unsigned (Reference to the loaded key) - digestAlgorithm - enum - SHA-256 - inputDataOffset - unsigned - inputDataLength - unsigned	- outputDataOffset - unsigned - outputDataLength - unsigned	- xxxxc / Invalid key -xxxxd / Invalid key type -xxxxf / Session to be reinitialized -xxxzg / Invalid Parameters -xxxxa / Operation failed -xxxxh - Authentication failure -xxxxl - Output buffer too small	Yes	Yes	No	Calculate the MAC / signature value and returns. keyId shall match with a loaded key compatible with the selected algorithm. Examples: - HMAC: secret symmetric key
11	Crypto	verify	- algorithm - enum - HMAC - keyId - unsigned (Reference to the loaded key) - digestAlgorithm - enum - SHA256 - inputDataOffset - unsigned - inputDataLength - unsigned - expectedSignatureOffset - unsigned - expectedSignatureLength - unsigned	bool - verification status	-xxxxh - Authentication failure -xxxxx / Invalid key -xxxxd / Invalid key type -xxxxy / Session to be reinitialized -xxxxz / Invalid Parameters -xxxxa / Operation failed	Yes	Yes	No	Calculate the MAC / signature value and verifies if the expected signature matches. Returns true if verification is successful or returns false. keyId shall match with a loaded key compatible with the selected algorithm. Examples: - HMAC: secret symmetric key
12	Crypto	encrypt	- cipherAlgorithm - enum - AES_CBC - keyId - unsigned (Reference to the loaded key) - initDataOffset - unsigned (optional) - initDataLength - unsigned (optional) - inputDataOffset - unsigned - inputDataLength - unsigned	- encryptDataOffset - unsigned - encryptDataLength - unsigned	-xxxxi - Invalid Init Data length -xxxxh - Authentication failure -xxxj - Input data size exceeded the supported limits -xxxxl - Output buffer too small -xxxxe - Invalid key -xxxxa - Operation failed	Yes	Yes	No	Returns the data after encryption. Input data to encrypt shall not exceed 1 MB of size. Length of encrypted data is returned as 0 on any errors. For AES_CBC algorithm: - Expects IV as initData of 16 bytes in length - While IV reuse may be allowed, Zero or NULL IV is not allowed. Recommended to use generateRandom API to generate random number. - inputData contains the buffer to be encrypted For other algorithms, in future: Nonce can be passed as initData as well.

13	Crypto	decrypt	- algorithm - enum - AES_CBC - keyId - unsigned (Reference to the loaded key) - initVectorOffset - unsigned - initVectorLength - unsigned - encryptDataOffset - unsigned - encryptDataLength - unsigned	- outputDataOffset - unsigned - outputDataLength - unsigned	-xxxxh - Authentication failure -xxxkj - Input data size exceeded the supported limits -xxxkl - Output buffer too small -xxxke - Invalid key -xxxxa - Operation failed	Yes	Yes	No	Returns the decrypted clear data. Input data to decrypt shall not exceed 1.5 MB of data. length of output data is returned as 0 on any errors For AES_CBC algorithm: - Expects IV as initData of 16 bytes in length - While IV reuse may be allowed, Zero or NULL IV is not allowed. Recommended to use generateRandom API to generate random number. - inputData contains the buffer to be encrypted
14	Crypto	deriveSessionKeys	- algorithm - enum - AUTH_DH - privateKeyId - unsigned - peerPublicKeyData - list of unsigned, ordered. - derivationKeyId - unsigned	- encryptionKeyId - unsigned - hmacKeyId - unsigned - wrappingKeyId - unsigned	-xxxxh - Authentication failure -xxxxa - Operation failed	Yes	Yes	No	Performs authenticated DH key derivation
15	Crypto	generateRandom	Number of bytes	randomValue - list of unsigned, ordered.	-xxxxh - Authentication failure -xxxxa - Operation failed	Yes	Yes	No	Returns cryptographically random data of specified length. Random value encoded is of length passed in as parameter